

Stabilizer Math — Complete Technical Formulation

Definitions (cycle n)

x_n : input/state at cycle n

ϵ : micro-factor (small positive number)

m_n : magnified value

r_n : fractional residual drift

α : fractional correction gain ($0 < \alpha < 1$)

1) Magnification

$$m_n = x_n / \epsilon$$

2) Residual

$$k_n = \text{round}(m_n)$$

$$r_n = m_n - k_n, \text{ with } r_n \in [-1/2, 1/2]$$

3) Fractional Correction

$$c_n = \alpha r_n$$

4) Apply Correction

$$m'_n = m_n - \alpha r_n = (1-\alpha)m_n + \alpha k_n$$

5) Restore Scale

$$x_{\{n+1\}} = \epsilon m'_n$$

Closed Form

$$x_{\{n+1\}} = (1-\alpha)x_n + \alpha \epsilon \text{round}(x_n / \epsilon)$$

Residual Dynamics

Assuming k constant locally:

$$r_{\{n+1\}} = (1-\alpha) r_n$$

$$|r_{\{n+1\}}| = |1-\alpha| |r_n|$$

Convergence Condition

$$|1-\alpha| < 1 \Rightarrow 0 < \alpha < 2 \text{ (monotonic for } 0 < \alpha < 1)$$

Convergence Rate

$$|r_N| = (1-\alpha)^N |r_0|$$

$$N \geq \ln(\tau/|r_0|) / \ln(1-\alpha)$$

Limit Point

$$\lim m_n = k \Rightarrow \lim x_n = \varepsilon k$$

Error in Original Units

$$e_n = x_n - \varepsilon k_n = \varepsilon r_n$$

$$e_{\{n+1\}} = (1-\alpha) e_n$$

Per-Cycle Reduction

$$\text{Reduction fraction} = \alpha$$

Every k Cycles

$$\text{Equivalent contraction } g = (1-\alpha)^{(1/k)}$$

$$\text{Average reduction} = 1 - g$$

Implementation Notes

ε must be representable numerically.

Rounding basin changes at half-integers (piecewise stability).

Define rounding explicitly (e.g., $\text{floor}(m+0.5)$).

Alternative attractors may replace $\text{round}()$ if grid locking is undesirable.